CIS 531 / 731: Generative AI Audit Log (GenAI.pdf)

Assignment: MP 5 (Structured Streaming Pipeline)
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Track: Prescriptive Analytics | Text Corpora | Track 2

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1. LLM Platform Utilized:
   - Tool: Gemini (via Generative AI Project Kit v2.3 Tollbooth workflow).
   - Interaction Date: October 20-21, 2026.

2. Scope of Generative Assistance:
   - Input Constraints Provided:
     * Course Enrollment: CIS 531 Undergraduate (Subject to policy INV-11: unmodified ontology schema).
     * Approved PS 4 Data Sourcing: Genius, MusicBrainz, Spotify, RateYourMusic.
     * Stream Parameters: Micro-batch file replay, chunk duration <= 30s, p95 lateness = 14.82s.
   - Tasks Synthesized:
     * Structuring deal contracts for BoundedChunking, Whitelist, and Provenance gates in contracts.py.
     * Drafting Structured Streaming pipeline skeleton (watermark delay calibration, window aggregations).
     * Formulating fault-injection harness and reconciliation procedure logic.

3. Verification and Integrity Check:
   - Policy INV-11 Compliance: Adopted instructor-provided CorpusChunk schema unmodified without custom inference.
   - Grounding & Validation: All deal contracts, Spark window definitions, and schema types were manually verified and debugged by the student.
   - Fault Tolerance Test: Successfully executed reconciliation.py on pinned CPU DevContainer baseline with verified state store recovery and zero duplicate window keys emitted.

4. Disclosure Statement:
   All conceptual assistance and code scaffolds provided by the AI Kit have been reviewed, tested, and verified against course infrastructure guidelines.